Building Technology

Engineering Mathematics I

Assignment 1 - Questions Only

Instructions: Answer all questions. Show formula, substitution, calculation, units and interpretation.

Section A: Formula and Concept Questions

1. Explain the meaning of the formula below and identify its main symbols. (4 marks)

Straight line: $y = mx + c$

2. Explain the meaning of the formula below and identify its main symbols. (4 marks)

$\tan \theta = \text{rise}/\text{run}$

3. Explain the meaning of the formula below and identify its main symbols. (4 marks)

Pyramid volume: $V = \frac{1}{3}Ah$

4. Explain the meaning of the formula below and identify its main symbols. (4 marks)

Circumference: $C = \pi d$

5. Explain the meaning of the formula below and identify its main symbols. (4 marks)

$dy/dx = f'(x)$

Section B: Applied Questions

Problem Set 1

6. Solve a building technology problem involving roof pitch. Show formula, substitution, correct units and a technical interpretation. (10 marks)

7. Solve a building technology problem involving stair gradients. Show formula, substitution, correct units and a technical interpretation. (10 marks)

8. Solve a building technology problem involving excavation volumes. Show formula, substitution, correct units and a technical interpretation. (10 marks)

9. Solve a building technology problem involving cost curves. Show formula, substitution, correct units and a technical interpretation. (10 marks)

10. Solve a building technology problem involving building survey data. Show formula, substitution, correct units and a technical interpretation. (10 marks)

Problem Set 2

11. Solve a building technology problem involving roof pitch. Show formula, substitution, correct units and a technical interpretation. (10 marks)

12. Solve a building technology problem involving stair gradients. Show formula, substitution, correct units and a technical interpretation. (10 marks)

13. Solve a building technology problem involving excavation volumes. Show formula, substitution, correct units and a technical interpretation. (10 marks)

14. Solve a building technology problem involving cost curves. Show formula, substitution, correct units and a technical interpretation. (10 marks)

15. Solve a building technology problem involving building survey data. Show formula, substitution, correct units and a technical interpretation. (10 marks)

Problem Set 3

16. Solve a building technology problem involving roof pitch. Show formula, substitution, correct units and a technical interpretation. (10 marks)

17. Solve a building technology problem involving stair gradients. Show formula, substitution, correct units and a technical interpretation. (10 marks)

18. Solve a building technology problem involving excavation volumes. Show formula, substitution, correct units and a technical interpretation. (10 marks)

19. Solve a building technology problem involving cost curves. Show formula, substitution, correct units and a technical interpretation. (10 marks)

20. Solve a building technology problem involving building survey data. Show formula, substitution, correct units and a technical interpretation. (10 marks)

Problem Set 4

21. Solve a building technology problem involving roof pitch. Show formula, substitution, correct units and a technical interpretation. (10 marks)

22. Solve a building technology problem involving stair gradients. Show formula, substitution, correct units and a technical interpretation. (10 marks)

23. Solve a building technology problem involving excavation volumes. Show formula, substitution, correct units and a technical interpretation. (10 marks)

24. Solve a building technology problem involving cost curves. Show formula, substitution, correct units and a technical interpretation. (10 marks)

25. Solve a building technology problem involving building survey data. Show formula, substitution, correct units and a technical interpretation. (10 marks)

References and Further Reading

These notes are original learning materials prepared for Mr Mwirigi Mathematics Training Hub. They are academically informed by standard engineering mathematics texts and the local TVET curriculum framework. No textbook pages or worked examples have been copied.

1. Stroud, K. A. and Booth, D. J. Engineering Mathematics, 8th edition. Bloomsbury Academic.

2. Bird, J. Engineering Mathematics. Routledge / Taylor & Francis.

3. Bird, J. Higher Engineering Mathematics. Routledge / Taylor & Francis.

4. Relevant Civil Engineering, Building Technology and Land Survey curriculum and occupational standard documents.